

Artificial Intelligence Systems II

Code: 8836556-01

Revision MCQ — With Answers

20 Questions per Lecture

Lecture 1 – Course Introduction & AI Overview

Q1. What is Artificial Intelligence (AI)?

- A) A programming language used to write software
- B) A field that builds systems to simulate or extend human intelligence in real-world tasks**
- C) A database management system
- D) A hardware device for fast computing

Correct Answer: B

Q2. Which of the following best defines Machine Learning?

- A) Hard-coding rules into a system manually
- B) Algorithms that learn from data to improve performance without explicit programming**
- C) A type of database query language
- D) A networking protocol for AI systems

Correct Answer: B

Q3. Deep Learning is best described as:

- A) A subset of ML that uses rule-based systems
- B) A completely separate field from Machine Learning
- C) A subset of ML that uses multi-layer neural networks to learn rich representations from raw data**
- D) A hardware optimization technique

Correct Answer: C

Q4. Which three core elements drive AI development?

- A) Rules, Experts, and Databases
- B) Computing Power, Data, and Algorithms**
- C) Hardware, Internet, and Cloud
- D) Sensors, Actuators, and Software

Correct Answer: B

Q5. What is the role of 'Data' in AI development?

- A) The engine of AI
- B) The brain of AI
- C) The powerhouse (fuel) of AI**
- D) The memory of AI

Correct Answer: C

Q6. What is the role of 'Algorithms' in AI development?

- A) The fuel of AI

B) The brain and command center of AI

C) The storage unit of AI

D) The engine of AI

Correct Answer: B

Q7. Natural Language Processing (NLP) is used in all of the following EXCEPT:

A) Chatbots and customer support

B) Machine translation (e.g., Google Translate)

C) Object detection in traffic monitoring

D) Sentiment analysis of text

Correct Answer: C

Q8. What does Computer Vision (CV) primarily aim to do?

A) Allow machines to 'see' and understand visual information from images or videos

B) Process and generate human language

C) Predict stock market prices

D) Optimize database queries

Correct Answer: A

Q9. Image Segmentation in Computer Vision creates:

A) Bounding boxes around objects

B) A pixel-level map partitioning the image into multiple segments

C) Text from scanned documents

D) 3D models from 2D images

Correct Answer: B

Q10. OCR (Optical Character Recognition) is used to:

A) Detect objects in images using bounding boxes

B) Translate images into different languages

C) Convert images of text into machine-encoded editable text

D) Segment images into regions

Correct Answer: C

Q11. Which course part covers 'Distributed training & model compression'?

A) Part 1: Classical ML Review

B) Part 2: Deep Learning Core

C) Part 3: Systems & Applications

D) Part 4: Final Exam

Correct Answer: C

Q12. Which limitation of Classical ML is mentioned as a motivation for Deep Learning?

- A) It requires too much computing power
- B) Heavy reliance on manual feature engineering and limited representation capacity**
- C) It cannot handle numerical data
- D) It always overfits to training data

Correct Answer: B

Q13. What does 'end-to-end trainable' mean in the context of deep learning?

- A) Training only the last layer of a neural network
- B) The model learns features and makes predictions all in one unified training process**
- C) Using a separate feature extractor before the model
- D) Training the model from the output layer backwards only

Correct Answer: B

Q14. Advanced NLP systems today are primarily based on:

- A) Rule-based grammar systems only
- B) Decision tree classifiers
- C) Large Language Models (LLMs) like BERT or ChatGPT**
- D) k-Nearest Neighbor classifiers

Correct Answer: C

Q15. Image Segmentation is vital in which of the following real-world applications?

- A) Email spam detection
- B) Medical imaging to isolate tumors and autonomous driving for road boundaries**
- C) Weather prediction models
- D) Speech recognition systems

Correct Answer: B

Q16. Which field specifically deals with giving machines the ability to 'see'?

- A) Natural Language Processing
- B) Reinforcement Learning
- C) Computer Vision**
- D) Recommender Systems

Correct Answer: C

Q17. Object Detection technologies can help with:

- A) Translating text from one language to another
- B) Detecting objects in an image for further processing with intelligent algorithms**
- C) Compressing large datasets
- D) Optimizing neural network weights

Correct Answer: B

Q18. Performance of Classical ML algorithms tends to:

- A) Always improve as more data is added
- B) Saturate with more data and struggle with high-dimensional raw data**
- C) Decrease with more data
- D) Be unaffected by the amount of training data

Correct Answer: B

Q19. Which of the following is NOT a limitation of classical ML mentioned in the course?

- A) Limited representation capacity
- B) Struggles with high-dimensional raw data like images and audio
- C) Performance saturates with more data
- D) Cannot perform classification tasks**

Correct Answer: D

Q20. The course covers which deep learning architectures?

- A) CNNs, RNNs, and Transformers only
- B) CNNs, RNNs, Transformers, BERT, DeepSeek and GPT**
- C) Decision Trees and Random Forests only
- D) SVMs and k-NN algorithms

Correct Answer: B

Lecture 2 – Review of Machine Learning Concepts

Q1. How is Machine Learning defined according to Tom Mitchell's classic definition?

- A) A program that follows explicit rules to solve problems
- B) A program that learns from experience E with respect to task T and improves performance P**
- C) A system that only works with numerical data
- D) A database querying method

Correct Answer: B

Q2. In the ML pipeline, what comes BEFORE model training?

- A) Model deployment
- B) Model evaluation
- C) Data preparation, cleansing, and feature selection**
- D) Feedback and iteration

Correct Answer: C

Q3. What is the 'curse of dimensionality'?

- A) The problem of having too few training samples
- B) What happens when you have too many features compared to the amount of data**
- C) Overfitting caused by deep neural networks
- D) The inability of a model to generalize

Correct Answer: B

Q4. Which type of ML uses labeled data with known correct outputs?

- A) Unsupervised Learning
- B) Reinforcement Learning
- C) Supervised Learning**
- D) Self-supervised Learning

Correct Answer: C

Q5. In Supervised Learning, the model improves by:

- A) Exploring the environment and receiving rewards
- B) Clustering similar data points together
- C) Comparing its predictions with correct labels and adjusting**
- D) Using unlabeled data to find hidden patterns

Correct Answer: C

Q6. Which of the following is an example of a Classification task?

- A) Predicting tomorrow's temperature

- B) Estimating next week's stock price
 - C) Deciding whether an email is spam or not**
 - D) Forecasting monthly sales revenue
- Correct Answer: C**
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Q7. Regression is used to predict:

- A) A category or class label
- B) Continuous numerical values**
- C) Cluster assignments
- D) Reward signals

Correct Answer: B

Q8. Unsupervised Learning works with:

- A) Labeled data with correct answers
- B) Unlabeled data; no correct answers are given**
- C) Data collected from environment interactions
- D) Data with partial labels

Correct Answer: B

Q9. In Reinforcement Learning, the agent learns by:

- A) Comparing its output to labeled data
- B) Grouping similar data points
- C) Interacting with the environment to maximize cumulative reward**
- D) Minimizing a regression loss function

Correct Answer: C

Q10. What is 'Overfitting' in machine learning?

- A) The training error is large, meaning the model is too simple
- B) The model performs well on training data but has large generalization error on new data**
- C) The model has equal training and test error
- D) The model is trained on too little data

Correct Answer: B

Q11. What is 'Underfitting'?

- A) The training error is small but generalization error is large
- B) The training error itself is large (model is too simple to capture the pattern)**
- C) The model has too many parameters
- D) The model memorizes training data

Correct Answer: B

Q12. What does 'Generalization' mean in ML?

- A) How well the model performs on training data
- B) The accuracy of predictions based on new, unseen data**
- C) The number of parameters in the model
- D) The speed of model training

Correct Answer: B

Q13. The Filter method for feature selection:

- A) Uses a prediction model to score feature subsets
- B) Evaluates each feature independently based on its correlation with the target variable**
- C) Treats feature selection as part of the modeling process
- D) Combines benefits of wrapper and embedded methods

Correct Answer: B

Q14. The Wrapper method for feature selection:

- A) Evaluates features independently without using a model
- B) Uses different combinations of features scored by a prediction model**
- C) Is the same as the Filter method
- D) Only works with neural networks

Correct Answer: B

Q15. Which feature selection method combines the benefits of both Filter and Wrapper methods?

- A) Filter method
- B) Wrapper method
- C) Embedded method**
- D) Greedy method

Correct Answer: C

Q16. What are the three common types of problems in dirty data?

- A) Missing, Duplicate, and Sorted data
- B) Incompleteness, Noise, and Inconsistency**
- C) Overfitting, Underfitting, and Bias
- D) Features, Labels, and Targets

Correct Answer: B

Q17. A 'Training Set' is used for:

- A) Evaluating the final generalization of the model
- B) Tuning hyperparameters only
- C) Building (learning) the model from data**
- D) Selecting the best model architecture

Correct Answer: C

Q18. A 'Test Set' is used for:

- A) Training the model weights
- B) Validating hyperparameter choices
- C) Evaluating the learned model's generalization capability**
- D) Feature engineering

Correct Answer: C

Q19. Why is Data Standardization important?

- A) To increase the number of features in the dataset
- B) To reduce noise and improve model accuracy by putting features on comparable scales**
- C) To remove all outliers from the dataset
- D) To convert categorical features into numerical ones

Correct Answer: B

Q20. When should Machine Learning be preferred over rule-based systems?

- A) When rules are simple and fixed
- B) When the problem is well-understood with stable rules
- C) When rules are complex, tasks change over time, or data distributions shift**
- D) When the dataset is very small

Correct Answer: C

Lecture 3 – Linear Regression and Decision Trees

Q1. What is a Model Parameter?

- A) A configuration set by the user before training
- B) A value learned from data that defines the model's function and is saved as part of the model**
- C) The number of training epochs
- D) The validation accuracy threshold

Correct Answer: B

Q2. Which of the following is a Model Hyperparameter?

- A) The weights of a neural network
- B) The bias terms in linear regression
- C) The learning rate and number of neurons in a neural network**
- D) The predicted output values

Correct Answer: C

Q3. In k-fold Cross-Validation, the dataset is:

- A) Split into two halves: training and testing
- B) Split into K equal parts, each used once as the test set while training on the remaining K-1 parts**
- C) Used entirely for training with a separate validation set
- D) Randomly shuffled K times without any structured splitting

Correct Answer: B

Q4. What is the main benefit of k-fold Cross-Validation?

- A) It increases the size of the training dataset
- B) It provides a more reliable and stable estimate of model performance**
- C) It eliminates the need for a test set entirely
- D) It automatically tunes hyperparameters

Correct Answer: B

Q5. Linear Regression assumes:

- A) A non-linear relationship between input and output
- B) A linear relationship between input (independent variable) and output (dependent variable)**
- C) Outputs must be binary (0 or 1)
- D) All input features must be categorical

Correct Answer: B

Q6. In the equation $y = b_0 + b_1 * x$, what does b_1 represent?

- A) The intercept of the regression line
- B) The dependent variable
- C) The slope (coefficient) that relates the input x to the output y**
- D) The error term (residual)

Correct Answer: C

Q7. What is a 'residual' in Linear Regression?

- A) The slope of the regression line
- B) The distance between the observed value and the value predicted by the regression line**
- C) The average of all predictions
- D) The number of data points used in training

Correct Answer: B

Q8. The Least Squares method minimizes:

- A) The sum of absolute differences between actual and predicted values
- B) The maximum single prediction error
- C) The sum of squared differences (residuals) between actual and predicted values**
- D) The total number of misclassified samples

Correct Answer: C

Q9. Mean Squared Error (MSE) is calculated as:

- A) The sum of all residuals
- B) The maximum residual value
- C) $(1/n) \times \Sigma(\text{predicted} - \text{actual})^2$**
- D) The square root of the sum of residuals

Correct Answer: C

Q10. Which of the following is a real-world application of Linear Regression?

- A) Classifying emails as spam or not spam
- B) Predicting salary based on years of experience**
- C) Clustering customers into groups
- D) Detecting objects in images

Correct Answer: B

Q11. In a Decision Tree, the 'Root Node' is:

- A) Any node that makes a final prediction
- B) A node with no children at the bottom of the tree
- C) The top node representing the full dataset and the first split**
- D) A node that performs no splitting

Correct Answer: C

Q12. What is a 'Leaf Node' in a Decision Tree?

- A) The first node at the top of the tree
- B) An internal node that performs a split
- C) A terminal node that makes the final prediction (no further splits)**
- D) A node with exactly two children

Correct Answer: C

Q13. How does a Decision Tree choose which feature to split on?

- A) It always splits on the first feature in the dataset
- B) It splits on the feature that results in the lowest Gini Impurity (most pure split)**
- C) It randomly selects a feature at each node
- D) It splits based on the highest feature value

Correct Answer: B

Q14. What does a Gini Impurity value of 0 mean?

- A) The node is maximally impure with equal class distribution
- B) The node is pure — all samples belong to one class**
- C) The node needs to be split further
- D) The feature was not useful for splitting

Correct Answer: B

Q15. What is the maximum Gini Impurity value for a binary classification problem?

- A) 0.0
- B) 0.25
- C) 0.5**
- D) 1.0

Correct Answer: C

Q16. Which of the following are valid Decision Tree splitting criteria mentioned in the lecture?

- A) MSE, MAE, and RMSE
- B) Gini Index, Information Gain, and Gain Ratio**
- C) Precision, Recall, and F1-Score
- D) Accuracy, Loss, and Entropy only

Correct Answer: B

Q17. A Decision Tree stops splitting when:

- A) The tree reaches exactly 10 nodes
- B) A node is pure, no features remain, or a stopping condition (e.g., max depth) is met**
- C) The training accuracy reaches 100%
- D) The test accuracy stops improving

Correct Answer: B

Q18. Decision Trees are described as 'upside down' because:

- A) Predictions flow from leaves to root
- B) The root node is at the top and leaves are at the bottom**
- C) They are read from right to left
- D) The deepest node is the most important

Correct Answer: B

Q19. The goal of each split in a Decision Tree is to:

- A) Maximize the number of nodes
- B) Minimize the depth of the tree
- C) Un-mix the data or reduce uncertainty at each node**
- D) Increase the Gini Impurity of child nodes

Correct Answer: C

Q20. Which variable is called the 'dependent variable' in Linear Regression?

- A) X — the input feature used for prediction
- B) b_0 — the intercept
- C) Y — the output variable being predicted**
- D) b_1 — the slope coefficient

Correct Answer: C

Lecture 4 – Random Forest and Ensemble Learning

Q1. What is the core idea of a Random Forest algorithm?

- A) Using a single very deep decision tree
- B) Using many decision trees, each trained on random data/features, with combined results**
- C) Applying gradient descent to decision tree nodes
- D) Training one tree on the full dataset multiple times

Correct Answer: B

Q2. How does Random Forest combine tree predictions for classification?

- A) Averaging the probability outputs of all trees
- B) Taking the prediction of the best-performing single tree
- C) Majority voting across all trees**
- D) Using the deepest tree's prediction

Correct Answer: C

Q3. How does Random Forest combine tree predictions for regression?

- A) Majority voting
- B) Averaging predictions across all trees**
- C) Using the median prediction
- D) Taking the maximum prediction value

Correct Answer: B

Q4. What is 'Bootstrapping' in Random Forests?

- A) Pruning trees to reduce their depth
- B) Taking random samples WITH replacement from the training data to create new datasets**
- C) Applying regularization to each tree
- D) Normalizing the feature values before training

Correct Answer: B

Q5. What makes Random Forest different from a single Decision Tree?

- A) Random Forest uses Gini Impurity while Decision Trees do not
- B) Random Forest introduces randomness by using random subsets of data and features for each tree**
- C) Random Forest is always less accurate than a single tree
- D) Random Forest only works for regression problems

Correct Answer: B

Q6. Which assumption of Random Forest states that results are more reliable with sufficient training data?

- A) Each tree makes its own decisions
- B) Random parts of the data are used
- C) Enough data is needed**
- D) All trees must agree on predictions

Correct Answer: C

Q7. Ensemble Learning solves the problem of a single model by:

- A) Using more training data than a single model
- B) Aggregating multiple models and balancing their errors**
- C) Training one model for a longer time
- D) Increasing the model's depth

Correct Answer: B

Q8. Which ensemble method trains models independently IN PARALLEL?

- A) Boosting
- B) Stacking's meta-learner phase
- C) Bagging**
- D) AdaBoost

Correct Answer: C

Q9. Which ensemble method trains models SEQUENTIALLY, each correcting the previous?

- A) Bagging
- B) Random Forest
- C) Boosting**
- D) Stacking's base model phase

Correct Answer: C

Q10. What is the main goal of Bagging?

- A) Reduce bias
- B) Reduce variance and prevent overfitting**
- C) Improve generalization through a meta-learner
- D) Focus on hard misclassified examples

Correct Answer: B

Q11. What is the main goal of Boosting?

- A) Reduce variance
- B) Reduce bias and turn weak learners into strong ones**
- C) Use entirely different model types
- D) Apply parallel training for efficiency

Correct Answer: B

Q12. In Boosting, after each iteration, misclassified samples receive:

- A) Lower weights so they are less likely to be selected
- B) The same weight as before
- C) Higher weights so subsequent models focus on them**
- D) Zero weight to be excluded from training

Correct Answer: C

Q13. Stacking combines predictions using:

- A) Simple majority voting
- B) Averaging all model predictions
- C) A meta-learner (second-level model) trained on base model outputs**
- D) Weighted voting based on model depth

Correct Answer: C

Q14. In Stacking, what type of models are typically used as base models?

- A) Only decision trees
- B) Only neural networks
- C) Different types of models (e.g., Trees, SVMs, KNN)**
- D) Only linear models

Correct Answer: C

Q15. What type of model is commonly used as the meta-learner in Stacking?

- A) A deep neural network
- B) Logistic Regression**
- C) A Random Forest
- D) A Support Vector Machine

Correct Answer: B

Q16. Which ensemble method uses the ENTIRE dataset but re-weights samples after each round?

- A) Bagging
- B) Stacking
- C) Random Forest
- D) Boosting**

Correct Answer: D

Q17. Can Bagging models be trained in parallel?

- A) No — they must be sequential like Boosting
- B) Yes — base models are trained independently so parallel training is possible**
- C) Only when using neural networks as base models
- D) Only when the dataset is very large

Correct Answer: B

Q18. Which of the following algorithms is an example of a Boosting method?

- A) Random Forest
- B) Bagged Trees
- C) AdaBoost and Gradient Boosting**
- D) Stacked Generalization

Correct Answer: C

Q19. Why do ensembles generally work better than single models?

- A) They use more complex loss functions
- B) They require more computational resources which improves accuracy
- C) Multiple models with different errors cancel each other out, reducing overall error**
- D) They always use larger datasets than single models

Correct Answer: C

Q20. In the Ensemble Models Summary, which method uses 'weighted sum based on model accuracy' for combination?

- A) Bagging
- B) Stacking
- C) Boosting**
- D) Random Forest

Correct Answer: C

Lectures 5 & 6 – Neural Networks: Architecture and Activation Functions

Q1. Neural Networks are described as computational models inspired by:

- A) The structure of computer processors
- B) How the human brain learns**
- C) Database management systems
- D) Rule-based expert systems

Correct Answer: B

Q2. Which of the following is a component of a Neural Network?

- A) Gini Index and information gain
- B) Neurons, Weights, Biases, and Activation Functions**
- C) Bootstrap samples and meta-learners
- D) Training sets and test sets only

Correct Answer: B

Q3. What are 'Weights and Biases' in a Neural Network?

- A) Fixed constants set by the data scientist
- B) Adjustable parameters that determine how strongly inputs affect a neuron**
- C) The output values of the network
- D) The activation functions applied at each layer

Correct Answer: B

Q4. The neural network training loop is:

- A) Data → Features → Model → Prediction
- B) Forward pass → Loss → Backward pass → Weight update**
- C) Input → Flatten → Softmax → Output
- D) Train → Test → Deploy → Monitor

Correct Answer: B

Q5. The Output Layer activation function for multi-class classification should be:

- A) ReLU
- B) Linear (identity)
- C) Tanh
- D) Softmax**

Correct Answer: D

Q6. The Output Layer activation function for binary classification should be:

- A) ReLU
- B) Sigmoid**
- C) Softmax
- D) Tanh

Correct Answer: B

Q7. The Output Layer activation function for regression tasks should be:

- A) Softmax
- B) Sigmoid
- C) Linear (identity)**
- D) ReLU

Correct Answer: C

Q8. Why are Activation Functions necessary in Neural Networks?

- A) To reduce the number of model parameters
- B) To initialize weights to zero
- C) To introduce non-linearity, enabling the network to learn complex patterns**
- D) To normalize the input data

Correct Answer: C

Q9. What happens if all layers use only Linear Activation Functions?

- A) The network becomes more powerful with more layers
- B) The entire network reduces to a single linear model, unable to learn non-linear patterns**
- C) The network trains faster and avoids overfitting
- D) The network automatically learns deep representations

Correct Answer: B

Q10. What is the output range of the Sigmoid activation function?

- A) $(-\infty, +\infty)$
- B) $(-1, 1)$
- C) $(0, 1)$**
- D) $(0, +\infty)$

Correct Answer: C

Q11. The Sigmoid activation function is associated with which major problem?

- A) Exploding gradients
- B) Dead neurons
- C) Vanishing gradient problem**
- D) Overfitting

Correct Answer: C

Q12. The Tanh activation function outputs values in the range:

- A) (0, 1)
- B) (-1, 1)**
- C) (0, $+\infty$)
- D) ($-\infty$, $+\infty$)

Correct Answer: B

Q13. Why is Tanh preferred over Sigmoid for hidden layers?

- A) Tanh is faster to compute than Sigmoid
- B) Tanh is zero-centered, which facilitates easier learning for subsequent layers**
- C) Tanh completely eliminates the vanishing gradient problem
- D) Tanh has a larger output range than Sigmoid

Correct Answer: B

Q14. How does ReLU (Rectified Linear Unit) function work?

- A) $f(x) = 1 / (1 + e^{-x})$
- B) $f(x) = (e^x - e^{-x}) / (e^x + e^{-x})$
- C) $f(x) = \max(0, x)$**
- D) $f(x) = x$ for all x

Correct Answer: C

Q15. What problem does Leaky ReLU solve compared to standard ReLU?

- A) The vanishing gradient problem for positive inputs
- B) The 'dying ReLU' problem where neurons output zero for all negative inputs**
- C) The exploding gradient problem
- D) The overfitting problem in deep networks

Correct Answer: B

Q16. The Softmax function is used to:

- A) Introduce non-linearity in hidden layers
- B) Replace ReLU in convolutional layers
- C) Transform output scores into a probability distribution summing to 1**
- D) Normalize input features

Correct Answer: C

Q17. According to the activation function comparison table, which function is used in 'Most hidden layers'?

- A) Sigmoid
- B) Tanh
- C) ReLU**
- D) Softmax

Correct Answer: C

Q18. Leaky ReLU differs from ReLU in that:

- A) It outputs 1 for all positive values
- B) It allows a small negative slope for negative inputs instead of outputting zero**
- C) It is used only in the output layer
- D) It has an output range of (0, 1)

Correct Answer: B

Q19. Which activation function gives 'much higher probability to the largest input value, while pushing all others closer to zero'?

- A) ReLU
- B) Sigmoid
- C) Tanh
- D) Softmax**

Correct Answer: D

Q20. The 'Learning Rule' in a neural network refers to:

- A) The choice of optimizer algorithm only
- B) How the network updates weights using error feedback to improve accuracy**
- C) The number of training epochs
- D) The activation function selection strategy

Correct Answer: B

Lecture 7 – Multilayer Perceptron (MLP) and Backpropagation

Q1. A Multilayer Perceptron (MLP) must have AT LEAST how many layers?

- A) One layer (input only)
- B) Two layers (input and output)
- C) Three layers (input, hidden, output)**
- D) Five layers

Correct Answer: C

Q2. The MLP is described as 'feedforward'. What does this mean?

- A) Information flows both forward and backward during prediction
- B) Information flows in one direction: Input → Hidden Layers → Output**
- C) Feedback loops exist between layers during inference
- D) The model trains backward from output to input

Correct Answer: B

Q3. Why can a Single-Layer Perceptron NOT solve the XOR problem?

- A) It does not have enough neurons
- B) It cannot handle more than two features
- C) XOR is not linearly separable — no single straight line can separate its classes**
- D) It uses the wrong activation function

Correct Answer: C

Q4. What does XOR output when both inputs are the same (e.g., 0,0 or 1,1)?

- A) 1
- B) 0**
- C) -1
- D) Undefined

Correct Answer: B

Q5. What does XOR output when the two inputs are different (e.g., 0,1 or 1,0)?

- A) 0
- B) -1
- C) 1**
- D) 0.5

Correct Answer: C

Q6. In Forward Propagation, the computation at each neuron is:

- A) Subtract inputs from weights and apply activation
- B) Compute the weighted sum of inputs plus bias, then apply an activation function**

- C) Multiply outputs of all previous layers together
- D) Apply the loss function directly to inputs

Correct Answer: B

Q7. What is the purpose of a Loss Function in a neural network?

- A) To decide the architecture of the network
- B) To measure how far the model's prediction is from the true label**
- C) To initialize the weight values
- D) To set the learning rate automatically

Correct Answer: B

Q8. Cross-Entropy Loss is most appropriate for:

- A) Regression problems predicting continuous values
- B) Clustering problems
- C) Classification tasks (works with Sigmoid for binary, Softmax for multi-class)**
- D) Dimensionality reduction tasks

Correct Answer: C

Q9. MSE (Mean Squared Error) is most appropriate for:

- A) Multi-class classification
- B) Binary classification
- C) Regression tasks predicting continuous values**
- D) Reinforcement learning reward calculation

Correct Answer: C

Q10. In Backpropagation, the error is propagated:

- A) From input to output, same direction as forward pass
- B) Only through the output layer
- C) From output back through the network layer by layer**
- D) Only to the first hidden layer

Correct Answer: C

Q11. What do gradients in backpropagation tell us?

- A) The optimal architecture for the next model
- B) How much each weight contributed to the error and the direction to reduce it**
- C) The number of training epochs required
- D) Which features to remove from the dataset

Correct Answer: B

Q12. Weights are updated in the direction that:

- A) Increases the loss function
- B) Is opposite to the gradient (reduces the error)**
- C) Maintains the current gradient value
- D) Increases the learning rate

Correct Answer: B

Q13. The Learning Rate controls:

- A) The number of hidden layers in the network
- B) How big each weight update step is during training**
- C) The choice of activation function
- D) The number of training samples used per epoch

Correct Answer: B

Q14. Does a standard MLP support feedback connections during prediction?

- A) Yes — MLP has recurrent connections in all layers
- B) Yes — feedback occurs between all adjacent layers
- C) No — MLP is feedforward only; information flows in one direction during prediction**
- D) Only in the output layer

Correct Answer: C

Q15. The formula $MSE = (1/n) \times \sum(\text{predicted} - \text{actual})^2$ measures:

- A) The maximum single prediction error
- B) The average of squared differences between predicted and actual values**
- C) The total absolute error across all predictions
- D) The variance of the model's predictions

Correct Answer: B

Q16. In backpropagation, calculating gradients requires:

- A) Only the output layer's values
- B) The derivative of the activation function and the chain rule**
- C) Re-running the forward pass from scratch
- D) The original training labels only

Correct Answer: B

Q17. What is the relationship between Forward Propagation and Backpropagation?

- A) They are the same process applied at different times
- B) Forward propagation makes a prediction; backpropagation learns from the mistake by updating weights**
- C) Backpropagation happens before forward propagation
- D) They are independent processes that don't interact

Correct Answer: B

Q18. Gradient Descent (SGD) is used during backpropagation to:

- A) Select the best model architecture
- B) Choose the correct activation function
- C) Optimize weights by moving in the direction that reduces the loss**
- D) Initialize the model parameters randomly

Correct Answer: C

Q19. The Hidden Layer(s) in an MLP:

- A) Directly receive the raw input data
- B) Produce the final class prediction
- C) Apply weighted sums and activation functions to transform the input representation**
- D) Only normalize the data before the output layer

Correct Answer: C

Q20. Which statement about the MLP is TRUE?

- A) MLP requires at least two hidden layers to function
- B) MLP can solve XOR because hidden layers and non-linear activations enable non-linear decision boundaries**
- C) MLP is the same as a single-layer perceptron with more neurons
- D) MLP cannot be trained using backpropagation

Correct Answer: B

Lecture 8 – CNN Basics: Convolution, Pooling, and Feature Extraction

Q1. Why is MLP not suitable for image classification on large images?

- A) MLP cannot use ReLU activation functions
- B) MLP ignores spatial relationships between pixels and causes a massive parameter explosion**
- C) MLP only handles grayscale images, not color images
- D) MLP cannot perform multi-class classification

Correct Answer: B

Q2. For a $224 \times 224 \times 3$ color image processed by an MLP, how many input neurons are required?

- A) 224
- B) 672
- C) 50,176
- D) 150,528**

Correct Answer: D

Q3. CNN addresses MLP's weakness through which three key concepts?

- A) Softmax, Cross-Entropy, and Gradient Descent
- B) Local Connectivity, Parameter Sharing, and Feature Extraction**
- C) Pooling, Flattening, and Fully-Connected Layers
- D) Batch Normalization, Dropout, and Regularization

Correct Answer: B

Q4. What is 'Local Connectivity' in CNNs?

- A) Each neuron connects to all pixels in the entire image
- B) Each neuron is connected only to a small local region of the input**
- C) Neurons are connected only to neurons in the next layer, not the previous
- D) All neurons share the same activation function

Correct Answer: B

Q5. What is 'Parameter Sharing' in CNNs?

- A) Different layers share different datasets
- B) The same filter/kernel (set of weights) is used across all spatial locations of the image**
- C) Parameters are shared between different network architectures
- D) Weights are shared between the encoder and decoder

Correct Answer: B

Q6. What does the Convolution operation produce?

- A) A probability distribution over classes
- B) A 1D vector ready for classification

- C) A Feature Map showing where a particular pattern was detected in the input**
D) A normalized version of the input image

Correct Answer: C

Q7. In a Feature Map, high values indicate:

- A) The filter did not match the pattern at that location
B) The location has zero activation
C) A strong match between the filter and the input at that location
D) The pooling operation reduced that region

Correct Answer: C

Q8. What activation function is most commonly applied after convolution in CNNs?

- A) Sigmoid
B) Tanh
C) Softmax
D) ReLU (Rectified Linear Unit)

Correct Answer: D

Q9. ReLU in CNNs keeps positive values and:

- A) Doubles all positive values
B) Replaces negative values with zero
C) Normalizes all values to the range (0, 1)
D) Converts all values to probabilities

Correct Answer: B

Q10. What does 'Stride' control in a CNN?

- A) The size of the convolutional filter
B) The number of filters applied to the input
C) How many pixels the kernel moves at each step across the input
D) The amount of padding added around the input

Correct Answer: C

Q11. A larger stride results in:

- A) A larger output feature map
B) A smaller output feature map
C) No change to the output size
D) A deeper network

Correct Answer: B

Q12. What is the purpose of Padding in CNNs?

- A) To add more training data
 - B) To prevent feature map size reduction and preserve border information during convolution**
 - C) To increase the number of filters
 - D) To apply dropout regularization
- Correct Answer: B**
-

Q13. What does the Pooling Layer do?

- A) It adds more learnable parameters to extract detailed features
 - B) It reduces the spatial dimensions of feature maps to improve efficiency and reduce overfitting**
 - C) It applies the activation function to raw pixel values
 - D) It connects feature maps to the output layer
- Correct Answer: B**
-

Q14. What is the purpose of the Flatten operation in a CNN?

- A) To reduce the number of feature maps
 - B) To apply max pooling across the spatial dimensions
 - C) To convert 2D or 3D feature maps into a 1D vector for the Fully-Connected layer**
 - D) To normalize activations before the output layer
- Correct Answer: C**
-

Q15. What does the Fully-Connected layer do in a CNN?

- A) Applies convolution filters to detect edges
 - B) Reduces spatial dimensions through pooling
 - C) Combines the learned feature representations to make the final prediction**
 - D) Applies ReLU activation to raw inputs
- Correct Answer: C**
-

Q16. What does the Output Layer of a CNN produce?

- A) A filtered version of the input image
 - B) Feature maps showing detected patterns
 - C) Final prediction scores or probabilities assigned to possible classes**
 - D) A 1D flattened vector of pixel values
- Correct Answer: C**
-

Q17. Early convolutional layers in a deep CNN learn:

- A) Complex abstract patterns like objects and faces
 - B) Simple low-level patterns such as edges and corners**
 - C) The probability distribution over all classes
 - D) Only the color information of the image
- Correct Answer: B**

Q18. Deeper convolutional layers in a CNN learn:

- A) Simple patterns like edges and textures
- B) Pixel-level color values
- C) More complex and abstract patterns like shapes and objects**
- D) The exact pixel locations of objects

Correct Answer: C

Q19. What is the correct order of operations in a typical CNN forward pass?

- A) Pooling → Convolution → ReLU → Flatten → Fully-Connected → Output
- B) Convolution → ReLU → Pooling → Flatten → Fully-Connected → Output**
- C) ReLU → Convolution → Flatten → Pooling → Output
- D) Flatten → Convolution → Pooling → ReLU → Output

Correct Answer: B

Q20. Compared to MLP, CNN preserves:

- A) More weights per neuron for richer representation
- B) The natural spatial structure of the input image**
- C) A fully connected architecture with all neurons linked
- D) More parameters in the final layers

Correct Answer: B

Lecture 9 – Classic CNN Architectures (LeNet-5, AlexNet, VGG-16)

Q1. Why does MLP fail for large image data compared to CNNs?

- A) MLP cannot use non-linear activation functions
- B) Flattening removes spatial structure and causes parameter explosion**
- C) MLP cannot perform batch training
- D) MLP requires GPU acceleration which is not available

Correct Answer: B

Q2. LeNet-5 was historically significant because:

- A) It introduced the Transformer architecture
- B) It was the first CNN to use attention mechanisms
- C) It demonstrated that CNNs can successfully solve digit recognition (MNIST)**
- D) It was the first model to achieve 99% accuracy on ImageNet

Correct Answer: C

Q3. LeNet-5 was designed primarily for:

- A) Large-scale object detection in natural images
- B) Medical image segmentation
- C) Handwritten digit recognition**
- D) Natural language processing tasks

Correct Answer: C

Q4. AlexNet was significant in deep learning history because:

- A) It introduced recurrent connections between layers
- B) It was the first model to use only 1×1 convolutions
- C) It won the ImageNet 2012 competition and showed deep CNNs trained on GPUs dramatically outperform traditional methods**
- D) It replaced pooling layers with attention mechanisms

Correct Answer: C

Q5. AlexNet was a 'breakthrough' primarily because it:

- A) Had fewer parameters than LeNet-5
- B) Made deep CNNs practical for large-scale image classification**
- C) Was the first model to use ReLU activations
- D) Introduced the concept of batch normalization

Correct Answer: B

Q6. What is the distinguishing architectural feature of VGG-16?

- A) Using large 11×11 and 5×5 convolutional kernels

- B) Using only 1×1 convolutions throughout the network
 - C) Stacking repeated 3×3 convolutions to build depth and learn features**
 - D) Using residual skip connections to avoid vanishing gradients
- Correct Answer: C**
-

Q7. Why does VGG-16 use 3×3 convolutions rather than larger kernels?

- A) 3×3 kernels are faster on CPUs only
- B) Stacking small 3×3 filters achieves larger receptive fields while keeping fewer parameters**
- C) Larger kernels cause the vanishing gradient problem
- D) 3×3 is the only kernel size supported by the framework used

Correct Answer: B

Q8. The name 'VGG-16' refers to:

- A) The year it was published (2016)
- B) The 16 types of filters used
- C) The 16 weight layers in the network**
- D) The 16 million parameters it uses

Correct Answer: C

Q9. In terms of depth, the correct ordering from shallowest to deepest is:

- A) VGG-16 $\rightarrow$ AlexNet $\rightarrow$ LeNet-5
- B) AlexNet $\rightarrow$ LeNet-5 $\rightarrow$ VGG-16
- C) LeNet-5 $\rightarrow$ AlexNet $\rightarrow$ VGG-16**
- D) LeNet-5 $\rightarrow$ VGG-16 $\rightarrow$ AlexNet

Correct Answer: C

Q10. Why did CNN architectures keep evolving after LeNet-5?

- A) LeNet-5 was too computationally expensive
- B) To become deeper, stronger, and more effective for increasingly complex tasks**
- C) Because CNNs were replaced by decision trees for a period
- D) Due to legal patents on the original LeNet architecture

Correct Answer: B

Q11. The key advantage of CNNs over MLPs for images is:

- A) CNNs have more parameters than MLPs
- B) CNNs preserve local spatial relationships and learn visual features efficiently**
- C) CNNs do not require backpropagation for training
- D) CNNs use fewer activation functions

Correct Answer: B

Q12. AlexNet introduced training on:

- A) CPUs using single-threaded computation
- B) GPUs to enable practical training of large deep networks**
- C) Distributed TPU clusters
- D) Quantum computing hardware

Correct Answer: B

Q13. What was the dataset used to measure AlexNet's breakthrough performance?

- A) MNIST (handwritten digits)
- B) CIFAR-10 (small images)
- C) ImageNet (large-scale image classification)**
- D) COCO (object detection)

Correct Answer: C

Q14. The main summary lesson from Lecture 9 states that CNN architectures evolved to become:

- A) Simpler, lighter, and less computationally demanding
- B) Deeper, stronger, and more effective**
- C) More reliant on manual feature engineering
- D) Less deep but wider with more filters per layer

Correct Answer: B

Q15. LeNet-5 used which type of data as input?

- A) High-resolution natural images (1000×1000)
- B) Text sequences for language modeling
- C) Small grayscale images of handwritten digits**
- D) Audio waveforms for speech recognition

Correct Answer: C

Q16. Which architecture first demonstrated that depth is critical for high-accuracy image recognition?

- A) LeNet-5
- B) AlexNet
- C) VGG-16
- D) Both AlexNet and VGG-16**

Correct Answer: D

Q17. Compared to an MLP, a CNN reduces parameters through:

- A) Using fewer neurons in fully-connected layers
- B) Local connectivity and parameter sharing via convolutional filters**
- C) Removing all hidden layers
- D) Using only the Softmax activation function

Correct Answer: B

Q18. What type of task were LeNet-5, AlexNet, and VGG-16 all primarily designed for?

- A) Object segmentation
- B) Image generation
- C) Image classification**
- D) Object detection with bounding boxes

Correct Answer: C

Q19. The progression from LeNet-5 to AlexNet to VGG-16 demonstrates:

- A) That simpler models always outperform complex ones
- B) That increasing depth and using appropriate architectural choices leads to better performance**
- C) That GPUs are not necessary for modern deep learning
- D) That convolution is no longer useful in modern architectures

Correct Answer: B

Q20. In all CNN architectures (LeNet-5, AlexNet, VGG-16), the final classification is performed by:

- A) A convolutional layer with Softmax
- B) The pooling layer
- C) Fully-connected layers leading to an output layer**
- D) A single neuron with sigmoid activation

Correct Answer: C

Lecture 11 – From Plain CNNs to ResNet and DenseNet

Q1. What is the primary problem that very deep plain CNNs suffer from?

- A) Too few parameters to learn complex features
- B) The vanishing/degrading gradient problem that prevents effective training**
- C) Excessive memory usage due to pooling layers
- D) Inability to use ReLU activation functions

Correct Answer: B

Q2. What is a 'residual connection' (skip connection) in ResNet?

- A) A connection that skips entire pooling layers
- B) A direct shortcut that adds the block's input to its output: $H(x) = F(x) + x$**
- C) A connection between the encoder and decoder
- D) A connection that multiplies input and output together

Correct Answer: B

Q3. The formula for a residual block is $H(x) = F(x) + x$. What does $F(x)$ represent?

- A) The original input to the block
- B) The identity mapping
- C) The residual mapping (what the layers inside the block learn)**
- D) The loss function output

Correct Answer: C

Q4. Why do skip connections help with the vanishing gradient problem?

- A) They reduce the number of layers gradients must flow through
- B) They allow gradients to flow directly through the shortcut path, bypassing intermediate layers**
- C) They increase the learning rate automatically
- D) They replace activation functions with linear ones

Correct Answer: B

Q5. ResNet stands for:

- A) Recursive Network
- B) Residual Network**
- C) Regression Network
- D) Redundant Network

Correct Answer: B

Q6. ResNet enabled training of very deep networks, reaching up to:

- A) 16 layers (like VGG-16)

- B) 8 layers (like AlexNet)
 - C) 100+ layers (e.g., ResNet-152)**
 - D) 50 neurons per layer
- Correct Answer: C**
-

Q7. Before ResNet, what happened when plain networks were made very deep?

- A) Accuracy always improved monotonically
- B) Training accuracy degraded — deeper models performed worse than shallower ones**
- C) The models became faster to train
- D) Overfitting completely disappeared

Correct Answer: B

Q8. What is the key idea that ResNet forces layers to learn?

- A) The full transformation $H(x)$ from scratch
- B) Only the residual (difference) $F(x) = H(x) - x$, relative to the identity mapping**
- C) The gradient of the loss function
- D) The pooling operation at each layer

Correct Answer: B

Q9. In DenseNet, each layer receives feature maps from:

- A) Only the immediately preceding layer
- B) Only the first layer of the network
- C) All preceding layers (dense connections)**
- D) Only layers of the same depth

Correct Answer: C

Q10. The main advantage of DenseNet's dense connectivity is:

- A) Reducing the number of layers needed
- B) Maximum feature reuse, reducing parameters while maintaining strong representation**
- C) Replacing residual connections with multiplicative connections
- D) Enabling bidirectional information flow

Correct Answer: B

Q11. How does DenseNet differ from ResNet in connectivity?

- A) DenseNet uses skip-one connections; ResNet connects every layer to every other
- B) ResNet connects each layer to every subsequent layer; DenseNet uses only skip-one
- C) DenseNet connects each layer to ALL subsequent layers; ResNet uses skip-one connections within blocks**
- D) They use identical connectivity patterns

Correct Answer: C

Q12. In ResNet's residual block, what happens if $F(x) = 0$?

- A) The block outputs zero and training stops
- B) The block performs the identity mapping $H(x) = x$ (passes input unchanged)**
- C) The block doubles the input value
- D) An error occurs and training diverges

Correct Answer: B

Q13. Which of these is a key benefit of ResNet over plain deep CNNs?

- A) ResNet requires no pooling layers
- B) ResNet uses larger kernels for better feature extraction
- C) ResNet can be trained much deeper (100+ layers) without performance degradation**
- D) ResNet eliminates the need for fully-connected layers

Correct Answer: C

Q14. DenseNet's parameter efficiency comes from:

- A) Using only 1×1 convolutions
- B) Feature reuse — each layer accesses features from all previous layers, reducing redundant learning**
- C) Removing batch normalization
- D) Using global average pooling instead of dense layers

Correct Answer: B

Q15. Residual connections in ResNet are added to the output of a block using:

- A) Multiplication of the input with the block output
- B) Concatenation (joining) of feature maps
- C) Element-wise addition of the input to the block output**
- D) Subtraction of the block output from the input

Correct Answer: C

Q16. DenseNet connects layers through:

- A) Element-wise addition like ResNet
- B) Concatenation of feature maps from all preceding layers**
- C) Weighted summation with learned coefficients
- D) Gating mechanisms similar to LSTM

Correct Answer: B

Q17. Which statement correctly distinguishes ResNet and DenseNet?

- A) ResNet uses concatenation; DenseNet uses addition
- B) ResNet uses element-wise addition in skip connections; DenseNet concatenates all previous feature maps**
- C) ResNet connects all layers; DenseNet uses only one skip connection per block

D) Both use identical connection patterns but with different activation functions

Correct Answer: B

Q18. The degradation problem in deep plain CNNs means:

A) Gradients become too large causing weight explosion

B) Adding more layers does NOT continue to improve (and can worsen) training accuracy

C) The model overfits because it has too many parameters

D) Pooling operations destroy important features

Correct Answer: B

Q19. Why is learning a residual mapping $F(x) = H(x) - x$ easier than learning $H(x)$ directly?

A) $F(x)$ always equals zero, simplifying computation

B) If the optimal function is close to identity, $F(x)$ is close to zero, which is easier to optimize than learning a full transformation

C) $F(x)$ does not require backpropagation

D) The residual mapping eliminates the need for activation functions

Correct Answer: B

Q20. Both ResNet and DenseNet were designed to solve problems encountered in:

A) Recurrent networks processing sequences

B) Training very deep convolutional networks

C) Attention-based transformer models

D) Shallow networks with few parameters

Correct Answer: B

Lecture 12 – Sequence Models: RNN and LSTM

Q1. Why are CNNs not well-suited for sequential data like text or speech?

- A) CNNs cannot use ReLU activations
- B) CNNs process inputs in parallel without maintaining memory of previous time steps**
- C) CNNs require too much GPU memory for sequential data
- D) CNNs cannot perform classification tasks

Correct Answer: B

Q2. What is the key feature of Recurrent Neural Networks (RNNs) that makes them suitable for sequences?

- A) They use convolutional filters to process sequences
- B) They maintain a hidden state that carries information from previous time steps**
- C) They process all time steps simultaneously in parallel
- D) They use attention mechanisms to weight input tokens

Correct Answer: B

Q3. In a vanilla RNN, the hidden state at time t is computed from:

- A) Only the current input x_t
- B) Only the previous hidden state h_{t-1}
- C) Both the current input x_t and the previous hidden state h_{t-1}**
- D) The future input x_{t+1} and current input x_t

Correct Answer: C

Q4. What is the major limitation of vanilla RNNs?

- A) They cannot process text data
- B) They suffer from the vanishing gradient problem, making it hard to learn long-term dependencies**
- C) They are too computationally efficient for complex tasks
- D) They require labeled data for all time steps

Correct Answer: B

Q5. LSTM stands for:

- A) Linear State Transfer Model
- B) Long Short-Term Memory**
- C) Layered Sequential Transformation Module
- D) Learned Sequence Transition Matrix

Correct Answer: B

Q6. What is the main purpose of the LSTM architecture?

- A) To process images faster than CNNs
 - B) To solve the vanishing gradient problem and enable learning of long-term dependencies**
 - C) To replace fully-connected layers in classification tasks
 - D) To perform attention-based sequence modeling
- Correct Answer: B**
-

Q7. An LSTM cell has how many gates?

- A) One (the output gate)
- B) Two (input and forget)
- C) Three (forget, input, and output)**
- D) Four (forget, input, update, output)

Correct Answer: C

Q8. What does the Forget Gate in an LSTM decide?

- A) What new information to add to the cell state
- B) What part of the output to expose from the cell state
- C) What information from the previous cell state to discard or keep**
- D) The learning rate for each time step

Correct Answer: C

Q9. What does the Input Gate in an LSTM decide?

- A) What to output from the cell state
- B) What irrelevant information to discard
- C) What new information from the current input to store in the cell state**
- D) The number of time steps to process

Correct Answer: C

Q10. What does the Output Gate in an LSTM decide?

- A) What to store in the cell state
- B) What information to forget from previous states
- C) What part of the cell state to expose as the hidden state output h_t**
- D) The final class prediction

Correct Answer: C

Q11. The cell state in an LSTM is often described as:

- A) The current input to the network
- B) The long-term memory that carries information across many time steps**
- C) The short-term hidden state updated at every time step
- D) The output probability distribution

Correct Answer: B

Q12. GRU stands for:

- A) Gated Recurrent Unit**
- B) General Regression Unit
- C) Gradient Recurrent Updater
- D) Global Recursive Unit

Correct Answer: A

Q13. How many gates does a GRU have?

- A) One
- B) Two (Update gate and Reset gate)**
- C) Three (same as LSTM)
- D) Four

Correct Answer: B

Q14. What is the main advantage of GRU over LSTM?

- A) GRU always achieves higher accuracy than LSTM
- B) GRU can process images unlike LSTM
- C) GRU is simpler with fewer parameters, making it faster to train while achieving comparable performance**
- D) GRU eliminates the vanishing gradient problem entirely

Correct Answer: C

Q15. The Update Gate in a GRU controls:

- A) How much new information from the current time step to add to the memory**
- B) What to output from the cell state
- C) The learning rate during backpropagation
- D) The forget and input gates simultaneously

Correct Answer: A

Q16. The Reset Gate in a GRU controls:

- A) What new information to store
- B) How much of the past hidden state to forget when computing the new hidden state**
- C) The output probability distribution
- D) The number of time steps to process

Correct Answer: B

Q17. RNNs are well-suited for which type of data?

- A) Grid-structured data like images
- B) Static tabular data with fixed features
- C) Sequential or time-series data where order and context matter**
- D) High-dimensional image datasets

Correct Answer: C

Q18. In an RNN, the vanishing gradient problem occurs because:

- A) The gradients become too large during backpropagation through time
- B) Gradients shrink exponentially as they are propagated back through many time steps**
- C) The model has too many parameters to optimize
- D) The hidden state grows too large to store

Correct Answer: B

Q19. Which architecture is a simplified version of LSTM that combines the cell state and hidden state?

- A) Vanilla RNN
- B) Bidirectional RNN
- C) GRU (Gated Recurrent Unit)**
- D) Deep RNN

Correct Answer: C

Q20. LSTM solves the long-term dependency problem through:

- A) Using larger hidden state vectors
- B) Processing sequences in reverse order
- C) A gated memory cell that selectively remembers or forgets information**
- D) Replacing recurrent connections with convolutional ones

Correct Answer: C

Lecture 13 – Attention and Transformers: From Sequence Modeling to GPT

Q1. What was the core limitation of RNNs/LSTMs that motivated the Attention mechanism?

- A) They were too fast to train, causing overfitting
- B) They compress all input information into a fixed-size context vector, losing information for long sequences**
- C) They could not be parallelized for image classification
- D) They required too much labeled data

Correct Answer: B

Q2. The Attention mechanism allows the model to:

- A) Process all tokens in parallel without any sequential computation
- B) Replace the need for training data
- C) Dynamically focus on different relevant parts of the input when generating each output token**
- D) Automatically tune the learning rate

Correct Answer: C

Q3. In the Attention framework, what does the Query (Q) represent?

- A) The content to be retrieved from memory
- B) What the model is currently looking for / asking about**
- C) All previous hidden states of the encoder
- D) The final output prediction

Correct Answer: B

Q4. In the Attention framework, what does the Key (K) represent?

- A) What the model is looking for
- B) The retrieved content used to form the output
- C) The representation of each input token that is matched against the Query**
- D) The gradient signal for weight updates

Correct Answer: C

Q5. In the Attention framework, what does the Value (V) represent?

- A) The query vector for the current position
- B) The actual content/information retrieved once the match ($Q \cdot K$) is computed**
- C) The position encoding added to the input
- D) The key used for alignment scoring

Correct Answer: B

Q6. Self-Attention allows:

- A) Each layer to attend to layers two steps below it
- B) The encoder to attend to the decoder's outputs
- C) Each token in a sequence to attend to all other tokens in the SAME sequence**
- D) The model to attend to training data during inference

Correct Answer: C

Q7. Why do Transformers require Positional Encoding?

- A) To reduce the number of attention heads
- B) To provide information about each token's position, since self-attention has no inherent sense of order**
- C) To normalize the key-value dot products
- D) To replace the embedding layer

Correct Answer: B

Q8. What is 'Multi-Head Attention'?

- A) Using one large attention head instead of many small ones
- B) Running multiple attention operations in parallel with different learned projections, then concatenating results**
- C) Applying attention only to the first and last token in each sequence
- D) A type of pooling applied to attention scores

Correct Answer: B

Q9. The Transformer architecture was introduced in the paper:

- A) 'ImageNet Classification with Deep CNNs' (AlexNet)
- B) 'Deep Residual Learning for Image Recognition' (ResNet)
- C) 'Attention Is All You Need'**
- D) 'Long Short-Term Memory'

Correct Answer: C

Q10. The standard Transformer architecture consists of:

- A) Only an Encoder (no Decoder)
- B) Only a Decoder (no Encoder)
- C) An Encoder stack and a Decoder stack connected by cross-attention**
- D) Convolutional layers with attention gates

Correct Answer: C

Q11. GPT (Generative Pre-trained Transformer) uses which part of the Transformer?

- A) The Encoder only
- B) Both Encoder and Decoder equally
- C) The Decoder only, with masked (causal) self-attention**

D) A hybrid CNN-Transformer encoder

Correct Answer: C

Q12. What is 'Masked Self-Attention' used in GPT?

A) Masking out the input embeddings during training

B) Preventing each token from attending to future tokens, enabling autoregressive generation

C) Hiding certain training samples from the model

D) Masking the gradient during backpropagation

Correct Answer: B

Q13. GPT generates text by:

A) Processing the entire sequence simultaneously and selecting the most likely sentence

B) Autoregressively predicting the next token given all previously generated tokens

C) Encoding the prompt and decoding a fixed-length response

D) Using beam search across all possible token combinations

Correct Answer: B

Q14. What key advantage do Transformers have over RNNs for processing long sequences?

A) Transformers use less memory than RNNs

B) Transformers process tokens sequentially like RNNs but with fewer parameters

C) Transformers process all tokens in parallel via self-attention, capturing long-range dependencies without sequential bottlenecks

D) Transformers do not require training data

Correct Answer: C

Q15. The attention score formula involves:

A) The dot product of the input embedding with itself

B) Computing $\text{softmax}(Q \cdot K^T / \sqrt{d})$ then multiplying by V

C) Multiplying the hidden state by the input embedding

D) Applying ReLU to the product of all token embeddings

Correct Answer: B

Q16. Why is the attention score scaled by $\sqrt{d}$ (square root of key dimension)?

A) To increase the magnitude of the dot products

B) To speed up computation

C) To prevent excessively large dot products that push the softmax into regions of very small gradients

D) To normalize the output probabilities to sum to 1

Correct Answer: C

Q17. BERT (Bidirectional Encoder Representations from Transformers) uses:

- A) Only the Transformer Decoder with causal masking
- B) Only the Transformer Encoder with bidirectional attention (attends to both past and future tokens)**
- C) A hybrid RNN-Transformer architecture
- D) Convolutional self-attention layers

Correct Answer: B

Q18. The Feed-Forward Network in a Transformer block:

- A) Applies convolutional filters to the attention output
- B) Applies two linear transformations with a non-linear activation (e.g., ReLU) applied position-wise**
- C) Replaces the self-attention mechanism in deeper layers
- D) Computes the positional encoding for each token

Correct Answer: B

Q19. In the Encoder-Decoder Transformer, the Decoder attends to the Encoder through:

- A) Self-Attention within the Decoder only
- B) Cross-Attention, where Decoder queries attend to Encoder keys and values**
- C) Pooling of Encoder outputs
- D) Concatenation of Encoder and Decoder hidden states

Correct Answer: B

Q20. Which statement best summarizes the Transformer's impact on AI?

- A) It replaced CNNs in all computer vision tasks
- B) It showed that recurrence is necessary for all sequence tasks
- C) Its parallel processing and attention mechanism enabled scaling to very large models (GPT, BERT), revolutionizing NLP and beyond**
- D) It eliminated the need for pre-training on large datasets

Correct Answer: C
